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A novel technique for microwave ablation of malignant pulmonary nodules: electromagnetic navigation bronchoscopy with real-time digital subtraction angiography and computed tomography imaging guidance

Yuan Xu^a, Qun Liu^b, Chao Guo^a, Huizhen Wang^c, Chenxi Ma^b, Tong Zhang^c, Yingzhi Qin^a, Hongsheng Liu^a, Zhijun Han^a, Naixin Liang^{a,*} and Shanqing Li^a

^aDepartment of Thoracic Surgery, Peking Union Medical College Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China

^bClinical Research Center for New Technology, Peking Union Medical College Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China

^cDepartment of Anesthesiology, Peking Union Medical College Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China

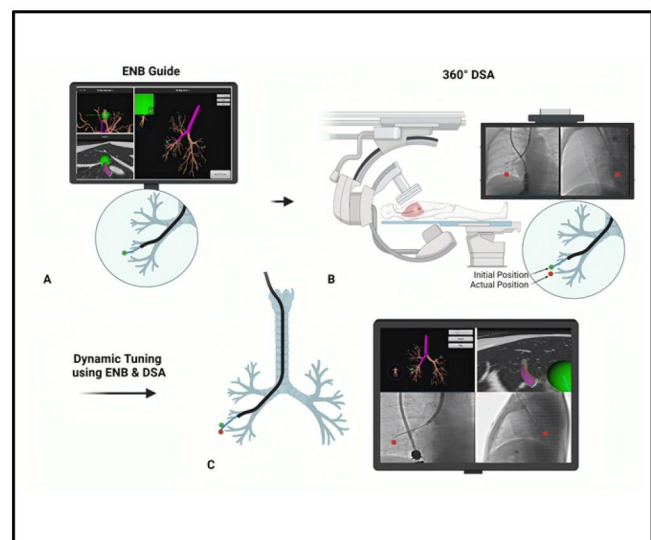
* Corresponding author. Department of Thoracic Surgery, Peking Union Medical College Hospital, 1# Shuaifuyuan, Dongcheng District, Beijing 100730, China. Tel: +86-15120003100; e-mail: pumchnelson@163.com (N. Liang).

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A Novel Technique for Microwave Ablation of Malignant Pulmonary Nodules: Electromagnetic Navigation Bronchoscopy with Real-time Digital Subtraction Angiography and Computed Tomography Imaging Guidance

Summary

In 39 patients with 51 malignant pulmonary nodules, we evaluated ENB-guided microwave ablation (MWA) with real-time DSA and 3D CBCT. Compared to traditional methods, this novel technique enhanced accuracy, reduced radiation exposure, and minimized complications, achieving 92.2% technical success with no major complications.



Legend: ENB – Electromagnetic Navigation Bronchoscopy; MWA – Microwave Ablation; DSA – Digital Subtraction Angiography; CBCT – Cone-Beam Computed Tomography

Abstract

OBJECTIVES: Electromagnetic navigation bronchoscopy (ENB) has been used to guide the diagnosis and treatment of pulmonary nodules. However, its combined application with real-time digital subtraction angiography (DSA), computed tomography (CT) and microwave ablation (MWA) for malignant pulmonary nodules has not been documented. This study aimed to evaluate the feasibility and safety of a novel technique that integrates ENB with real-time two-dimensional (2D) and three-dimensional (3D) imaging guidance for MWA of malignant pulmonary nodules.

METHODS: Thirty-nine patients with 51 malignant pulmonary nodules underwent MWA using ENB with real-time DSA and cone-beam CT (CBCT). The 2D DSA was used to guide the positioning of the ablation probe in real-time, while 3D CBCT was performed before and after ablation to confirm the probe position and assess the ablation zone. The primary outcomes were technical success and complications. Secondary outcomes included navigation time, procedure time, radiation exposure and ablation parameters.

RESULTS: The ENB-guided MWA with real-time 2D and 3D imaging guidance was technically successful in 92.2% (47/51) nodules. The median navigation time was 12 min, and the median total procedure time was 85 min. The median total radiation dose from the CBCT scans was 276 mGy. No major complications occurred. Minor complications included haemoptysis ($n = 2$) and postoperative fever ($n = 3$), which resolved spontaneously.

CONCLUSIONS: ENB combined with real-time 2D DSA guidance and 3D CBCT may be a feasible and safe technique for MWA of malignant pulmonary nodules.

Keywords: Electromagnetic navigation bronchoscopy • Cone-beam computed tomography • Microwave ablation • Pulmonary nodules • Fusion guidance

ABBREVIATIONS

2D	Two-dimensional
3D	Three-dimensional
CBCT	Cone-beam Computed tomography
CT	Computed tomography
DSA	Digital subtraction angiography
ENB	Electromagnetic navigation bronchoscopy
MWA	Microwave ablation
NCCN	National Comprehensive Cancer Network
RPak	Reference point air kerma
SBRT	Stereotactic body radiation therapy

INTRODUCTION

Pulmonary nodules, especially those with malignant potential, pose significant diagnostic and therapeutic challenges. With the widespread use of low-dose computed tomography screening, the detection rate of early-stage lung cancer has increased dramatically [1]. However, accurately localizing and treating these nodules while minimizing invasiveness and complications remains a critical clinical challenge.

Electromagnetic navigation bronchoscopy (ENB) has emerged as a promising approach for the diagnosis and treatment of pulmonary nodules, including early-stage lung cancer [2]. ENB uses electromagnetic fields to guide a steerable probe through the bronchial tree to the target lesion, enabling biopsy, fiducial marker placement or ablation [3]. However, the accuracy of ENB is affected by various factors, especially the respiratory movement of the nodules [4].

Cone-beam computed tomography (CBCT) provides effective three-dimensional (3D) imaging guidance during procedures. Recent reports on the fusion of ENB with CBCT have demonstrated promising accuracy [5]. However, this approach has notable limitations: first, CBCT cannot provide real-time guidance, thus hindering dynamic adjustments during the procedure; second, repeated CBCT scans increase the patient's exposure to radiation. These limitations have restricted the widespread

adoption of CBCT, and reports on the use of ENB-CBCT fusion guidance for the ablation of pulmonary nodules are quite limited.

In this study, we present a novel technique that integrates ENB with real-time digital subtraction angiography (DSA) and 3D CBCT for the microwave ablation (MWA) of malignant pulmonary nodules. This real-time integration not only reduces the need for repeated computed tomography (CT), thereby minimizing radiation exposure, but also allows for dynamic adjustments during the procedure, enhancing overall treatment precision. To the best of our knowledge, this is the first report on the use of this innovative fusion guidance technique, combining ENB with real-time DSA and 3D CBCT imaging, for the treatment of malignant pulmonary nodules.

METHODS

Patients

This study enrolled 39 patients with malignant pulmonary nodules who underwent MWA at Peking Union Medical College Hospital between January and November 2024. The inclusion criteria were: (i) nodules with a diameter ≤ 3 cm; (ii) nodules suspected to be malignant based on imaging features; and (iii) patients deemed medically inoperable, refusing surgery or suitable for a hybrid surgical approach (i.e. resection of some nodules and ablation of others). The exclusion criteria were: (i) benign nodules and (ii) severe cardiopulmonary dysfunction. This study was approved by the Peking Union Medical College Hospital Ethics Committee (I-22PJ1133), and all patients provided written informed consent.

Procedures

Preoperative planning. A chest CT scan was performed within 1 month before the procedure (Fig. 2A). The CT images were imported into the SuperDimension™ navigation system (Medtronic, USA) for procedural planning, including target definition, pathway selection and registration (Fig. 1).

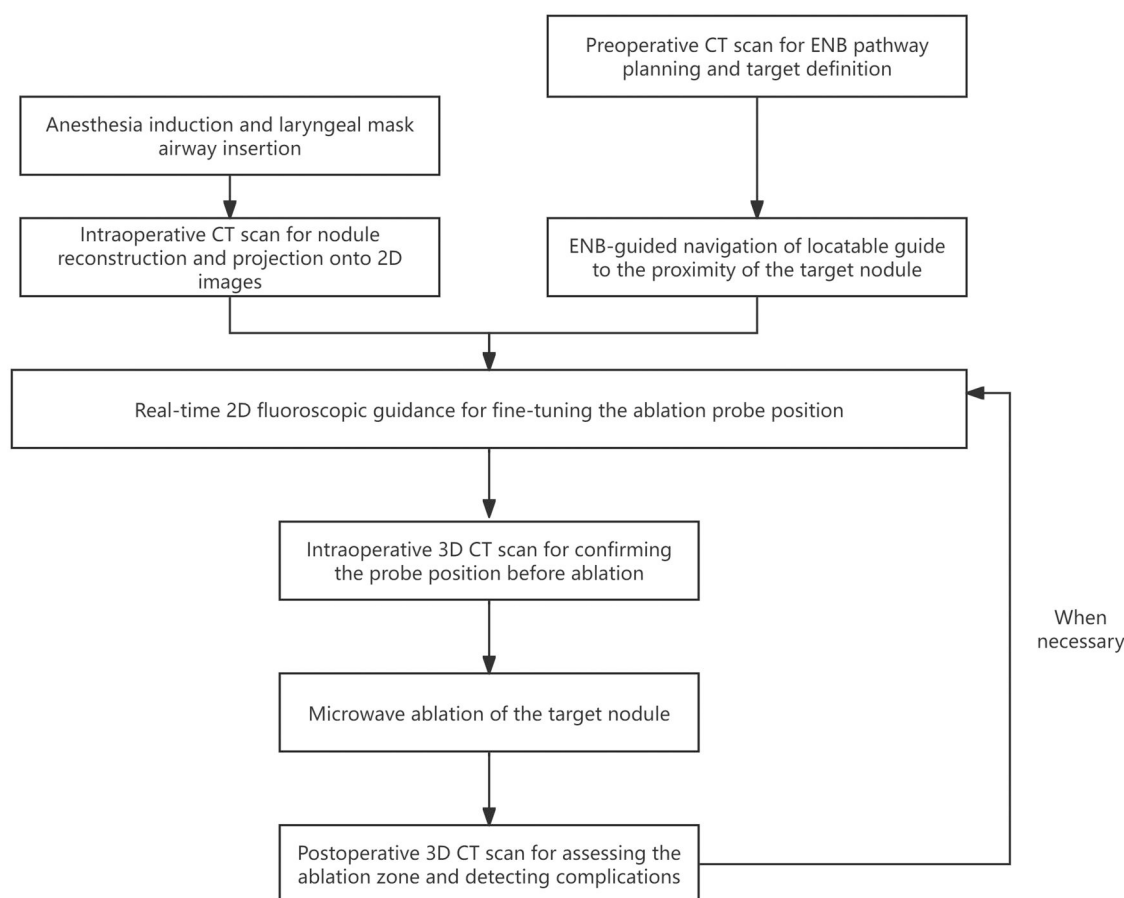


Figure 1: Procedural workflow of the novel technique integrating ENB with real-time 2D DSA guidance and 3D CT for microwave ablation of malignant pulmonary nodules. 2D: two-dimensional; 3D: three-dimensional; CT: computed tomography; DSA: digital subtraction angiography; ENB: electromagnetic navigation bronchoscopy.

Anaesthesia and ventilation. General anaesthesia was induced, and a laryngeal mask airway was inserted.

Intraoperative computed tomography scanning and nodule reconstruction. The patient was placed in a supine position on the operating table. An intraoperative CT scan (GE Discovery IGS 70R; 3DCT scan mode, rotation speed 16/s, 200° rotation) was performed to visualize the pulmonary nodule. The nodule was segmented and reconstructed in 3D using the AW Volume Viewer software. The reconstructed nodule information, including its location, size and shape, was then imported into the real-time DSA system.

Electromagnetic navigation bronchoscopy navigation. A locatable guide was inserted through the bronchoscope working channel and navigated to the target nodule under ENB guidance (Fig. 2B).

Real-time digital subtraction angiography guidance. Once the locatable guide was close to the nodule, real-time DSA was performed to confirm its position. If the position was not satisfactory, real-time dynamic tuning of the locatable guide was made under the fusion guidance of DSA and ENB, utilizing the 360° view provided by DSA to achieve the desired position at

the edge of the nodule (Fig. 2C and D). The tuning method is shown in Fig. 3. Once the locatable guide was in the desired position, it was withdrawn and replaced with the ablation probe (MTC-3CA-II37, Nanjing Weijing). If a biopsy was required, the locatable guide could also be replaced with biopsy forceps. The position of the ablation probe or biopsy forceps was then reconfirmed using real-time DSA.

Intraoperative three-dimensional computed tomography confirmation. An intraoperative 3D CT scan was performed to confirm the accurate positioning of the ablation probe within the nodule (Fig. 2E).

Ablation. MWA was carried out at 40–65W for 5–10 min, with adjustments made based on the nodule size and probe position. Multiple overlapping ablations were performed when necessary to ensure adequate ablation zone coverage.

Post-ablation evaluation. A CBCT scan was performed after ablation to evaluate the ablation zone coverage. Complications, if any, were recorded. Chest CT was performed on postoperative day 1 to assess technical success, defined as the ablation zone covering the nodule (Fig. 2F). The radiation dose recorded is the reference point air kerma (RPAK), which is provided by the CBCT system.

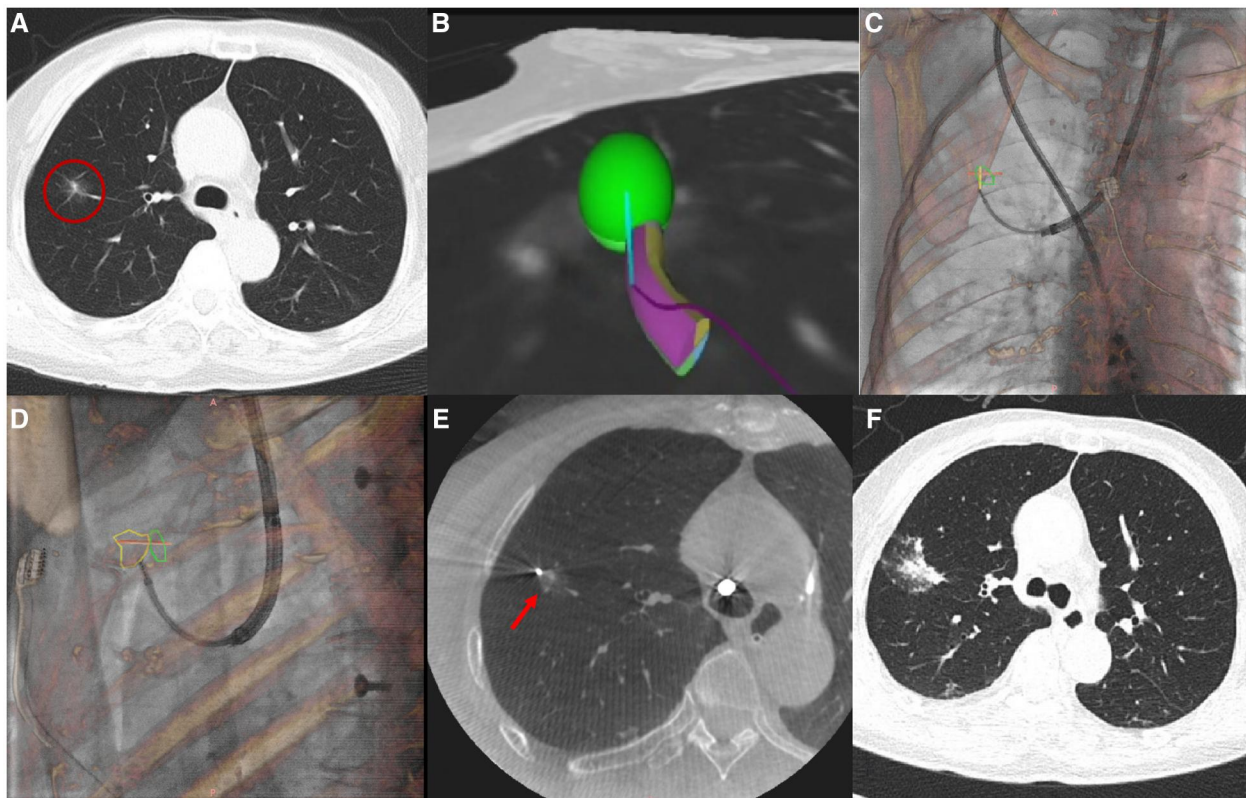


Figure 2: A representative case of ENB-guided microwave ablation with real-time 2D and 3D imaging guidance. (A) Preoperative CT scan showing a malignant pulmonary nodule (circle). (B) ENB image demonstrating the locatable guide reaching the target nodule under navigation guidance. (C and D) Real-time 2D DSA images demonstrating the positioning of the locatable guide. The nodule's maximum cross-sectional area is marked by orange, yellow and green circles in the axial, coronal and sagittal planes, respectively. (E) Initial DSA image showing the locatable guide just reaching the nodule. (F) DSA image after fine-tuning the position of the locatable guide, ensuring optimal placement for subsequent biopsy or ablation probe insertion. (G) Axial CT image demonstrating the accurate positioning of the ablation probe (arrow) adjacent to the nodule. (H) Assessment of follow-up CT image obtained on the 1st postoperative day, confirming adequate coverage of the nodule by the ablation zone. 2D: two-dimensional; 3D: three-dimensional; CT: computed tomography; DSA: digital subtraction angiography; ENB: electromagnetic navigation bronchoscopy.

Data analysis

Continuous variables were summarized as median (range) due to the potential non-normal distribution of the data. Categorical variables were presented as frequencies and percentages. Given the descriptive nature of this study, no inferential statistical analyses were performed.

RESULTS

Patient characteristics

The baseline characteristics of the patients are listed in Table 1. The median age of the 39 patients (7 males, 32 females) was 61 years. The median diameter of the target nodules was 12 mm. A total of 51 nodules were involved. Representative cases demonstrating the rationale for selecting MWA over surgical resection are shown in Fig. 4.

Procedural outcomes

The ENB-CBCT fusion guidance procedure was technically successful in all the cases. The median navigation time from the tip of the locatable guide to the closest position to the nodule was

12 min (range, 6–32 min). The median number of CBCT scans performed during the procedure was 3 (range, 2–7).

After the initial navigation, the median distance between the locatable guide tip and the nodule edge was 10 mm (range, 0–25 mm). With the aid of real-time DSA-CBCT fusion guidance, the ablation antenna was successfully positioned at the nodule edge in all cases, with a median error of 1.0 mm (range, 0–5 mm).

The median ablation time was 12 min (range, 5–24 min). The number of ablation sessions performed for each nodule ranged from 1 to 4, with a median of 2 sessions. The median total procedure time, from the start of navigation to the end of ablation and the final CBCT confirmation, was 85 min (range, 36–146 min). The median total radiation dose from the CBCT scans was 276 mGy (range, 107–913 mGy).

Technical success and complications

Technical success, defined as a coverage of the nodule by the ablation zone on the immediate post-ablation CBCT and post-operative day 1 chest CT, was achieved in 47/51 nodules (92.2%). Complete coverage was observed in 47 nodules, while the ablation zone partially covered the nodule in 4 nodules, with potential for complete coverage observed in subsequent follow-ups (usually 1–3 months). No major complications, such as

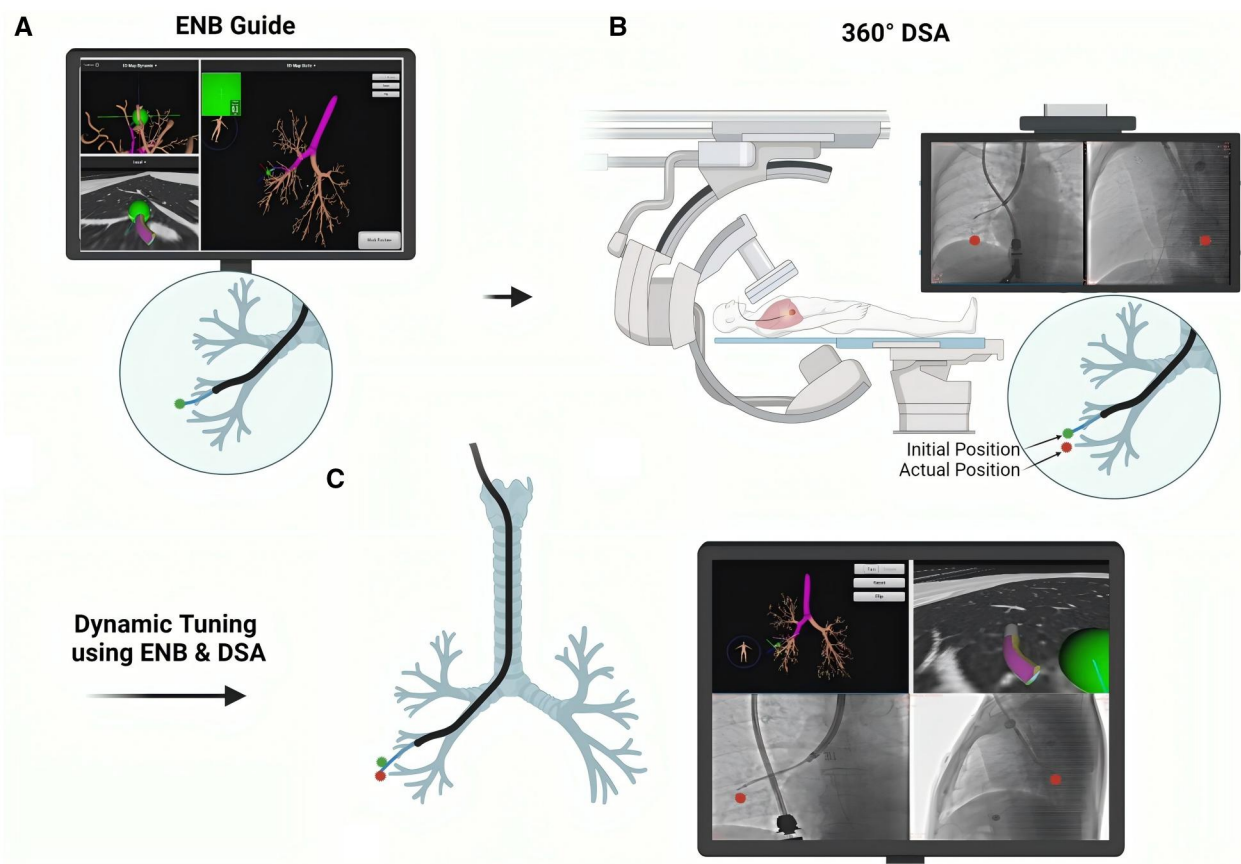


Figure 3: Schematic illustration of the dynamic real-time tuning of the locatable guide using the fusion of ENB and DSA. **(A)** Under ENB guidance, the locatable guide is navigated to the target location in the right lower lobe (nodule marked by green dot). **(B)** 360° DSA reveals that the actual nodule position is slightly anterior and inferior to the initial target location, as shown in the anteroposterior and lateral views (nodule marked by red dot). **(C)** With the real-time feedback from DSA, the locatable guide is dynamically adjusted towards the anterior and inferior direction under ENB guidance to compensate for the initial targeting error. After the dynamic tuning process, the locatable guide reaches the precise location. DSA: digital subtraction angiography; ENB: electromagnetic navigation bronchoscopy.

Table 1: Clinical characteristics of the total cohort		
Characteristic		No
N		39
Gender, n (%)	Male	7 (17.9%)
	Female	32 (82.1%)
Age, median (range)		61 (35–86)
Nodule features, n (%)	Pure GGO	30 (58.8%)
	Part-solid	16 (31.4%)
	Solid	5 (9.8%)
Nodule size (mm), median (range)		12 (6–26)
Involved lobe, n (%)	Right upper lobe	13 (25.5%)
	Right middle lobe	5 (9.8%)
	Right lower lobe	8 (15.7%)
	Left upper lobe	14 (27.5%)
	Left lower lobe	8 (15.7%)
Comorbidities, n (%)	Hypertension	10 (25.6%)
	Diabetes	5 (12.8%)
	Coronary artery disease	7 (17.9%)
	Other cancers	7 (17.9%)
Pulmonary function	FEV1 (median, range)	2.14 (0.99–4.03)
	FEV1% (median, range)	96.7 (46.5–123)

FEV1: forced expiratory volume in 1 s; GGO: ground-glass opacity.

pneumothorax, haemoptysis or death occurred. Minor complications included haemoptysis ($n=2$) and postoperative fever ($n=3$). They were all observed on the first postoperative day and resolved spontaneously without further intervention.

Due to the indolent nature of early-stage lung cancer and the limited follow-up duration, long-term outcomes such as local tumour progression and survival were not assessed in this initial report. These outcomes will be evaluated and reported in future studies with longer follow-up periods.

DISCUSSION

The present study introduces a novel technique that integrates ENB with real-time DSA and 3D CT for the MWA of malignant pulmonary nodules. This fusion navigation technique represents a significant advancement in the precision and safety of pulmonary nodule ablation. To the best of our knowledge, this is the first report of such a fusion navigation technique for the MWA of malignant pulmonary nodules.

Compared to traditional percutaneous CT-guided ablation, ENB-guided ablation has the advantage of lower complication

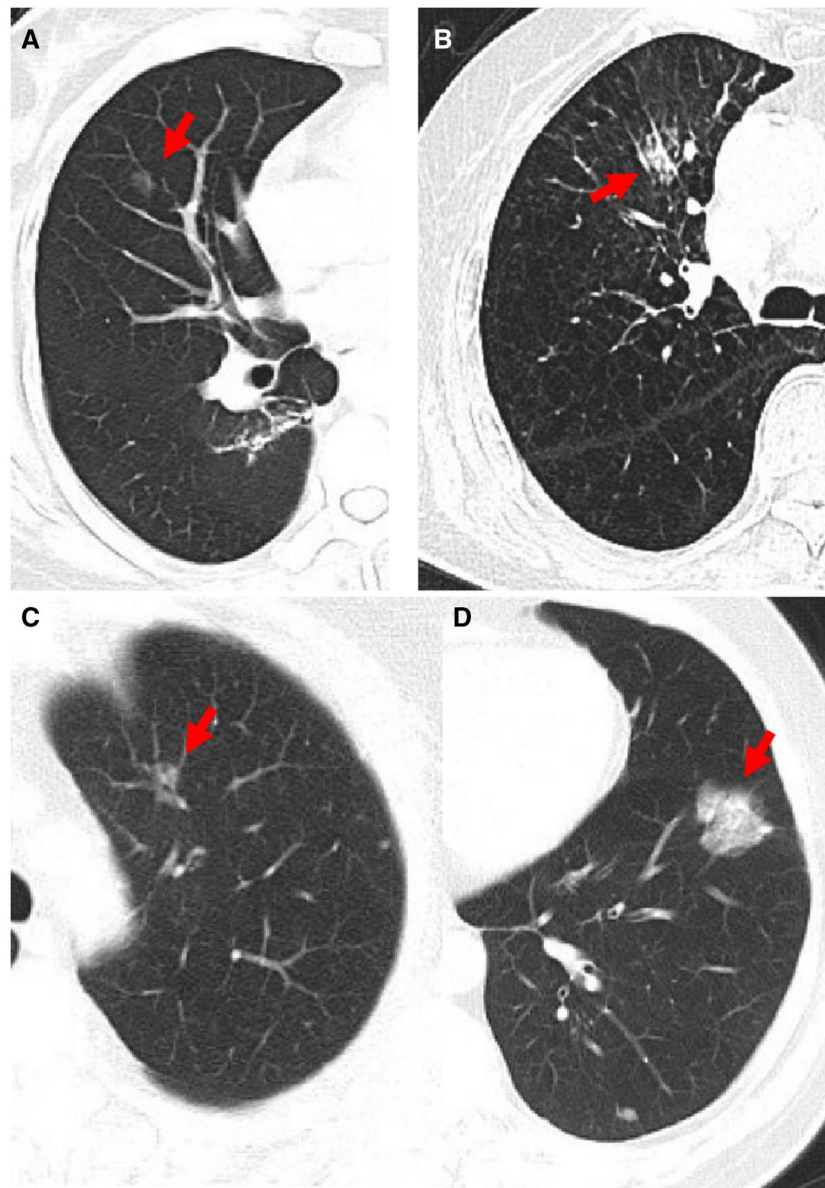


Figure 4: Representative cases demonstrating the rationale for selecting microwave ablation (MWA) over surgical resection. **(A)** A 53-year-old female patient with multiple primary lung cancer (MPLC) and a history of right upper lobectomy (pathology: adenocarcinoma, T1aN0M0). During follow-up, a growing ground-glass opacity (GGO) was detected in the right middle lobe, located in a deep position (arrow). **(B)** An 86-year-old female patient with MPLC. The right upper lobe lesion showed significant growth during follow-up (arrow). The patient had poor lung function (FEV1 0.86), a history of coronary artery disease and a previous coronary stent implantation. The nodule was located in a deep position. **(C, D)** A 63-year-old female patient with MPLC. Two lesions in the left upper lobe and left lower lobe showed growth during follow-up (arrows). A hybrid surgical strategy was adopted: the larger lesion in the left lower lobe was selected for surgical resection, while the smaller and deeper lesion in the left upper lobe was chosen for ablation.

rates, particularly pneumothorax [6], especially for deep-seated nodules. However, the accuracy of ENB-guided ablation has been a concern due to navigation error and the respiratory movement of the nodules [4, 7]. The introduction of CBCT has improved the accuracy of ENB-guided procedures by providing real-time 3D imaging guidance [5, 8]. Nevertheless, the repeated adjustment of the ablation probe under CBCT guidance can lead to increased radiation exposure [9]. Our innovative fusion technique addresses these challenges by combining the benefits of ENB, real-time two-dimensional (2D) DSA and 3D CT. We can perform dynamic real-time tuning of the locatable guide using the combination of ENB and DSA. As demonstrated in Fig. 3, this

innovative approach allows for precise adjustment of the guide position to compensate for any initial targeting errors detected by the 360° DSA imaging. Compared to traditional CT-guided blind adjustment, which often requires multiple scans and prolonged radiation exposure, our fusion method enables quick, real-time and accurate tuning with minimal radiation dose. The real-time feedback provided by DSA ensures that the locatable guide reaches the optimal position adjacent to the target nodule, enhancing the precision of subsequent biopsy or ablation procedures. Furthermore, the 3D CT scans performed before and after ablation provide crucial information for confirming the probe position and assessing the ablation zone. This streamlined

workflow not only ensures accuracy but also reduces radiation exposure and procedure time [10].

We believe that this innovative approach represents a future trend in the precise and safe management of pulmonary nodules, offering a promising alternative for patients who are inoperable or refuse surgery. Although the evidence level for ablation is not as high as that for surgery, this minimally invasive approach offers a promising alternative for several patient subgroups. First, patients who cannot tolerate surgery or have poor general conditions, such as the elderly, those with impaired cardiopulmonary function or those with severe comorbidities, may benefit from this technique. Second, patients with limited lung function reserve due to various reasons, such as previous partial lung resection, who are unable to tolerate further lung surgery, may benefit from this approach. Third, in cases of multiple pulmonary nodules that cannot be completely resected through surgery, a hybrid surgical approach involving various treatment modalities, including ablation, may be employed. Finally, this technique may be particularly useful for nodules located in deep positions or close to important anatomical structures, where surgical resection would result in significant loss of lung function or high risk. As demonstrated in the representative cases shown in Fig. 4, our novel fusion navigation MWA technique can effectively address the challenges associated with treating these specific patient subgroups. By combining the advantages of ENB, real-time 2D DSA and 3D CT, we can precisely target and ablate deep-seated nodules while minimizing the risk of complications and preserving lung function.

The radiation dose reported in this study is the RPAK provided by the CBCT system. It is important to note that RPAK does not directly reflect the actual radiation dose absorbed by the patient, as it does not consider the attenuation or scattering of X-rays within the patient's body. The actual absorbed dose is typically lower than the RPAK value. Therefore, the radiation dose values presented in this study should be interpreted as a reference for comparing the radiation exposure levels among different imaging protocols, rather than a direct measure of the patient's absorbed dose. Compared to conventional CT-guided MWA, which often requires multiple CT scans for probe positioning and treatment assessment, our technique minimizes the number of CT scans needed by using real-time 2D DSA guidance. This novel approach has the potential to significantly reduce the patient's overall radiation exposure.

Stereotactic body radiation therapy (SBRT) is another well-established local treatment option for lung nodules. The National Comprehensive Cancer Network (NCCN) guidelines for NSCLC recommend SBRT as a primary treatment option for patients with medically inoperable early-stage NSCLC or those who decline surgery, both for single and multiple lung nodules [11]. In recent years, ablation has gained increasing support from evidence-based medicine and is now presented as a viable alternative to SBRT for multiple lung nodules in the NCCN guidelines [11]. In clinical practice, when recommending local treatment for lung nodules in patients who are not suitable for surgical resection, both SBRT and ablation are discussed as potential options. The choice between SBRT and ablation often depends on individual patient characteristics, preferences and institutional expertise [12]. Future studies directly comparing the efficacy, safety and patient preferences between SBRT and ablation for the

treatment of lung nodules would provide valuable insights to guide clinical decision-making.

Limitations

This study had several limitations. The sample size was relatively small, and the follow-up period was short. Future studies with larger patient cohorts and longer follow-up are needed to further validate the long-term efficacy and safety of this technique. As a technical feasibility study, our primary goal was to demonstrate the successful application of our novel approach and to report key outcomes that highlight its potential benefits. Future studies with larger sample sizes and formal power calculations will be necessary to confirm the statistical significance of our findings and to further validate the efficacy of this technique in the management of malignant lung nodules. Another limitation of our study is its single-centre design, which may limit the generalizability of our findings to other healthcare settings. To further validate the effectiveness and reproducibility of our novel technique, multicentre studies will be necessary.

CONCLUSION

In conclusion, our results suggest that integrating ENB with real-time 2D DSA and 3D CBCT is a technically feasible and safe approach for lung nodule localization and ablation. The innovative integration of these technologies offers a promising new approach for patients who are inoperable or refuse surgery.

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Conflict of interest: none declared.

DATA AVAILABILITY

The data are available from the corresponding author on reasonable request.

ETHICAL STATEMENT

The authors are accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. This study was approved by the Ethics Committee of Peking Union Medical College Hospital (I-22PJ1133) and conducted in accordance with the Declaration of Helsinki (as revised in 2013).

Author contributions

Yuan Xu: Conceptualization; Data curation; Writing—original draft; Writing—review & editing. **Qun Liu:** Project administration; Writing—review & editing. **Chao Guo:** Project administration. **Huizhen Wang:** Project administration; Writing—review & editing. **Chenxi Ma:** Methodology; Writing—review & editing. **Tong Zhang:** Methodology. **Yingzhi Qin:** Methodology; Writing—review

& editing. **Hongsheng Liu:** Methodology. **Zhijun Han:** Methodology. **Naixin Liang:** Conceptualization; Writing–review & editing. **Shanqing Li:** Project administration.

Reviewer information

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